Research Methodology
Course Progress 44.72% (11 of 12)
Incremental vs Transformational Research and Technology : Part 02
Imagination in Research
Quiz: Week 10 : Assignment
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The Role of AI in Research
Introduction to Generative AI
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Assessment

Week 11 : Assignment

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-09-29, 14:09 IST.

Due date on 2026-10-07, 23:59 IST

- [ ] The machine was explicitly programmed with every possible checkers strategy.
- [x] The machine learned through experience by repeatedly playing the game.
- [x] The example is associated with an early instance of reinforcement learning.
- [ ] The example demonstrates that AI-based learning is a phenomenon that began only with generative AI.

Yes, the answer is correct. Score: 1

Accepted answer is The machine learned through experience by repeatedly playing the game., The example is associated with an early instance of reinforcement learning.

2. Which of the following statements correctly reflect the lecture's discussion of classical AI and generative AI? 1 Point

- [ ] Generative AI is the only appropriate approach for solving modern research problems.
- [x] Classical AI methods can still provide effective solutions for domain-specific problems.
- [x] Researchers should select an AI approach according to the problem rather than assuming that generative AI must always be used.
- [ ] Classical AI became irrelevant after the emergence of LLMs.

Yes, the answer is correct. Score: 1

Accepted answer Classical AI methods can still provide effective solutions for domain-specific problems., Researchers should select an AI approach according to the problem rather than assuming that generative

Yes, the answer is correct. Score: 1

Accepted answer is Classical AI methods can still provide effective solutions for domain-specific problems., Researchers should select an AI approach according to the problem rather than assuming that generative AI must always be used.

3. Consider an LLM that predicts the next token from the context provided to it. Which of the following statements are correct? 1 Point

- [x] The probability of a token can depend on the preceding context.
- [ ] A token with the highest probability must always be generated.
- [x] Different contexts can result in different probability distributions for the next token.
- [x] The model can repeatedly predict tokens to construct a longer response.

Yes, the answer is correct. Score: 1

Accepted answer is The probability of a token can depend on the preceding context., Different contexts can result in different probability distributions for the next token., The model can repeatedly predict tokens to construct a longer response.

4. Which of the following correctly distinguish token size, context size, and parameter count in an LLM? 1 Point

- [x] Token size is related to the vocabulary of words or subwords that the model can generate.
- [x] Context size refers to the amount of previous token information considered by the model.
- [ ] Parameter count refers to the number of tokens that can be stored in the prompt.
- [x] Parameter count is related to how the learned probability distributions are represented in the model.

Yes, the answer is correct. Score: 1

Accepted answer is Token size is related to the vocabulary of words or subwords that the model can generate., Context size refers to the amount of previous token information considered by the model., Parameter count is related to how the learned probability distributions are represented in the model.

5. Which statements correctly describe the role of vector representations in the LLM framework explained in the lecture? 1 Point

- [x] A very large probability matrix can be represented through a comparatively smaller-dimensional vector space.
- [x] Different contexts may be projected to nearby locations in the vector space.
- [x] Such compression can contribute to variability and errors because information from different contexts may influence generation.
- [ ] Every possible context is retained as a completely independent point with no overlap.

Yes, the answer is correct. Score: 1

Accepted answer is A very large probability matrix can be represented through a comparatively smaller-dimensional vector space., Different contexts may be projected to nearby locations in the vector space., Such compression can contribute to variability and errors because information from different contexts may influence generation.

6. Which of the following statements are consistent with the lecture's explanation of alignment? 1 Point

- [x] Alignment can modify model behaviour so that generated probabilities better correspond to desired outcomes.
- [x] Alignment may be directed towards objectives such as factuality, bias reduction or local context.
- [ ] Alignment guarantees that all outputs produced by an LLM are factually correct.
- [x] Reinforcement learning from human preferences is associated with the alignment process.

Yes, the answer is correct. Score: 1

Accepted answer is: Alignment can modify model behaviour so that generated probabilities better correspond to desired outcomes., Alignment may be directed towards objectives such as factuality, bias reduction or local context., Reinforcement learning from human preferences is associated with the alignment process.

7. **A researcher asks a small language model a question whose meaning is ambiguous. Relevant information is retrieved from an internal repository and added to the original question before generating the answer. Which statements correctly describe this approach?** 1 Point

- [x] The approach is consistent with Retrieval-Augmented Generation (RAG).
- [x] The retrieved information provides additional context to the model.
- [ ] The approach necessarily requires retraining the entire language model.
- [x] Additional context can make the model's response more focused.

Yes, the answer is correct. Score: 1

Accepted answer is: The approach is consistent with Retrieval-Augmented Generation (RAG)., The retrieved information provides additional context to the model., Additional context can make the model's response more focused.

8. **The lecture uses the example of the 2011 ICC World Cup to explain the effect of retrieval-augmented generation. Which of the following correctly captures the idea?** 1 Point

- [x] Additional retrieved information can contextualize an otherwise ambiguous query.
- [x] The original question can be effectively shifted toward a more appropriate contextual representation.
- [ ] RAG works primarily by increasing the number of parameters in the LLM.
- [x] A richer context can increase the probability of producing the intended

Yes, the answer is correct. Score: 1

Accepted answer is: Additional retrieved information can contextualize an otherwise ambiguous query., The original question can be effectively shifted toward a more appropriate contextual representation., A richer context can increase the probability of producing the intended

9. **Which of the following statements about hallucination are correct according to the lecture?** 1 Point

- [x] A model may produce a plausible answer even when the generated information does not match the ground truth.
- [x] Hallucination can occur when the model has insufficiently represented information or context.
- [x] Hallucination can be reduced through techniques such as providing larger context, but it cannot necessarily be eliminated completely.
- [ ] Hallucination means that the model has deliberately decided to provide false information.

Yes, the answer is correct. Score: 1

Accepted answer is: A model may produce a plausible answer even when the generated information does not match the ground truth., Hallucination can occur when the model has insufficiently represented information or context., Hallucination can be reduced through techniques such as providing larger context, but it cannot necessarily be eliminated completely.

10. **A language model produces a misleading recommendation because a false or deliberately misleading article was present in information from which the model learned. Which statements are appropriate?** 1 Point

- [ ] The problem is necessarily an example of hallucination.
- [x] AI systems can potentially propagate misinformation present in their information sources.
- [x] A response may be misleading even when it is not simply an invented fact by the model.
- [x] Independent verification of important AI-generated information remains necessary.

Yes, the answer is correct. Score: 1

Accepted answer is: AI systems can potentially propagate misinformation present in their information sources., A response may be misleading even when it is not simply an invented fact by the model., Independent verification of important AI-generated information remains necessary.

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